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Escuela Profesional de Agronomía
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Prof. Dr. Leslie A. Weston
Editor-in-Chief
Agronomy (MDPI)

Dear Editor,

I hereby enclose the manuscript entitled "Multi-trait selection of promising quinoa (*Chenopodium quinoa* Willd.) accessions in an Andean valley of Ayacucho, Peru", by Cerda Tenorio et al., to be considered for publication as an original research article in *Agronomy*.

In this study, we characterized the phenotypic diversity of 144 quinoa accessions from the germplasm bank of the Universidad Nacional de San Cristóbal de Huamanga, using 28 qualitative and 13 quantitative descriptors. We then compared five multi-trait selection methods, principal component analysis, the Elston and Mulamba–Mock indices, ideotype distance, and Random Forest, to prioritize accessions based on grain yield, thousand-seed weight, plant height, and stem diameter. Although the selection scores were positively correlated among methods, each method favored different trait profiles, revealing a trade-off between grain yield and seed size. By applying a consensus criterion, we identified twelve accessions selected by at least three methods as promising candidates for multienvironment validation and breeding programs. To our knowledge, this is one of the few studies to compare classical selection indices with machine learning for the prioritization of Andean quinoa germplasm, offering a practical framework for breeders and genebank curators.

We believe this manuscript is suitable for Agronomy, as it addresses germplasm evaluation and multitrait selection for crop improvement. The manuscript has not been published and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission.

Yours sincerely,

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